

Week 05: International Research Funding: Horizon Europe, ERC and Beyond

R&D and Project Management in Computer Science

R&D and Project Management in Computer Science (11117BLG002). Prof. Dr. Utku Kose. Full Professor, Department of Computer Engineering, Süleyman Demirel University, Isparta, Türkiye. Additional affiliations: University of North Dakota (USA), Universidad Panamericana (Mexico City, Mexico), Vel Tech University (Chennai, India). ORCID 0000-0002-9652-6415. E-mail: utkukose@sdu.edu.tr, utku.kose@und.edu, ukose@up.edu.mx, utkukose@gmail.com. This course is updated in line with current developments in the field. Last update: September 2026.

Overview

International programmes offer larger budgets, international partners and visibility, at the price of intense competition and detailed rules. This week explains the structure of Horizon Europe, its action types, funding rates and evaluation criteria, the grants of the European Research Council and other international instruments [1, 2, 3].

Learning outcomes

By the end of the week, students are expected to describe the pillars and action types of Horizon Europe, to calculate the EU contribution to a consortium budget, to apply the evaluation thresholds and ranking rules, to state the main features of ERC grants, and to identify other international instruments such as COST, EUREKA and national foundations abroad.

Lecture

Horizon Europe

Horizon Europe, the European Union's research and innovation programme for 2021 to 2027, has three pillars: Excellent Science, which includes the European Research Council and the Marie Skłodowska-Curie Actions; Global Challenges and European Industrial Competitiveness, organised in thematic clusters; and Innovative Europe, which includes the European Innovation Council. Türkiye is an associated country, so organisations in Türkiye can take part in its calls. Most collaborative calls take one of three forms [2]. Research and innovation actions produce new knowledge or explore new technologies. Innovation actions are closer to the market and produce plans, prototypes and demonstrations. Coordination and support actions fund networking, coordination and dissemination rather than research.

Funding rates and budgets

Research and innovation actions and coordination and support actions are funded at 100 percent of eligible costs. Innovation actions are funded at 70 percent, except for non-profit legal entities, which receive 100 percent [2, 4]. Indirect costs are covered by a flat rate of 25 percent of the eligible direct costs, with exclusions such as subcontracting. Figure 5.1 shows the consequence: 100 euro of direct costs yield 125 euro of eligible costs, of which a non-profit partner receives 125 euro and a for-profit partner in an innovation action 87.50 euro [4]. The lab and the notebook compute these amounts for a whole consortium.

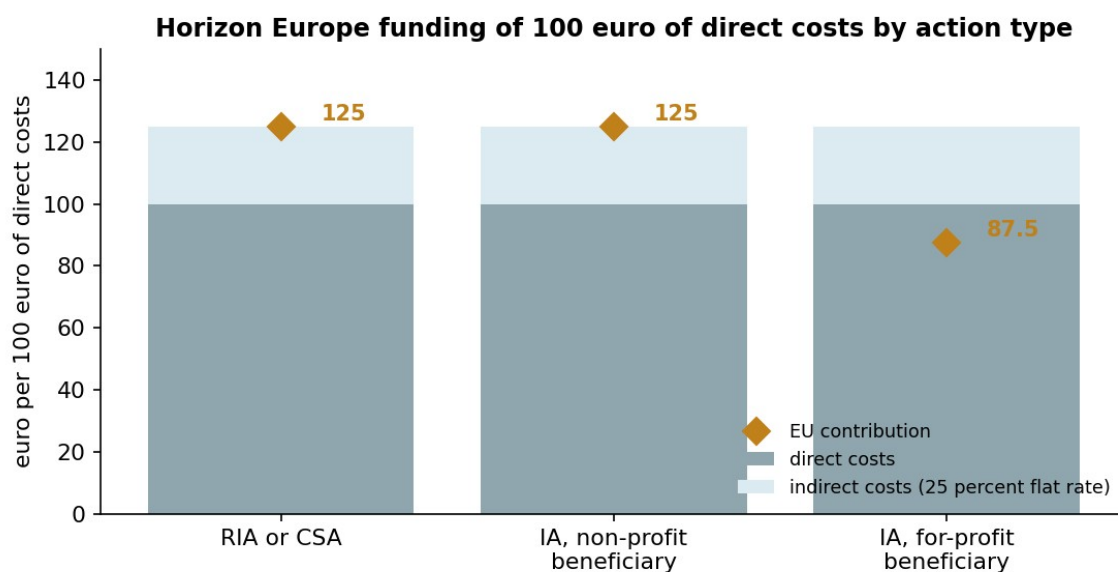


Figure 5.1. Eligible costs and EU contribution for 100 euro of direct costs in Horizon Europe, by action type and type of beneficiary [2, 4].

Evaluation

Collaborative proposals are scored by independent experts on three criteria: excellence, impact, and the quality and efficiency of the implementation [1]. Each criterion is scored from 0 to 5. By default, a proposal must reach at least 3 on each criterion and at least 10 in total to be considered for funding, unless the work programme states otherwise. Scores are normally not weighted, but for innovation actions the impact criterion receives a weight of 1.5 when proposals are ranked, not when thresholds are checked. Page limits are set in the proposal template of each call and were changed for the 2026 and 2027 work programmes, so the template attached to the call is the only authoritative source.

The ERC and other instruments

The European Research Council funds investigator-driven frontier research with a single evaluation criterion, scientific excellence. For the 2026 calls, Starting Grants of up to 1.5 million euro for five years address researchers 2 to 7 years after the doctorate, Consolidator Grants of up to 2 million euro address those 7 to 12 years after it, and Advanced Grants of up to

2.5 million euro address established leaders [3]. Additional funding of up to 1 million euro can be requested, and up to 2 million euro for researchers moving to Europe from elsewhere. From 2027, the Starting Grant window will run from the doctorate to 10 years after it, and the Consolidator window from 5 to 15 years [5].

Other instruments serve other purposes. COST Actions fund research networks rather than research itself. EUREKA and its Eurostars programme support market-oriented projects of companies, with national funding that TÜBİTAK provides through its 1509 programme. Outside Europe, funders apply their own criteria: The National Science Foundation of the United States, for example, reviews proposals for intellectual merit and broader impacts [6].

Pause and reflect. Would your project idea fit a research and innovation action, an innovation action or an ERC grant? What would have to change for it to fit another instrument?

Hands-on components

Interactive lab: Part A computes eligible costs and the EU contribution for a three-partner consortium. Part B applies the thresholds and the ranking rule. Part C compares ERC eligibility windows under the 2026 and 2027 rules [1, 2, 5]. Link: <https://utkukose.github.io/rd-project-management-NB-lecture/weeks/week-05/lab.html>

Colab notebook: The notebook computes the EU contribution to a Horizon Europe consortium with the official funding rates and flat rate for indirect costs, and applies the thresholds and ranking rule to a set of scored proposals [1, 2]. Link: https://colab.research.google.com/github/utkukose/rd-project-management-NB-lecture/blob/main/weeks/week-05/NB05_international_funding.ipynb

Reflection prompts

1. Which Horizon Europe action type suits your project idea, and what does that imply for your partners?
2. How would a 70 percent funding rate change the decision of a company to join a consortium?
3. Which international instrument could help you build a network before a full proposal?

Weekly task and submission

Draft a one-page concept note for an international call of your choice: the call and action type, the consortium with roles, a budget with the EU contribution computed as in the notebook, and a paragraph explaining how the proposal will address each evaluation criterion. Cite the official work programme and attach the notebook with both exercises completed.

The weekly task supports self-learning and builds a personal portfolio. When the course is followed with the instructor during an active semester, the task can be sent together with the exported learning log to utkukose@sdu.edu.tr or utkukose@gmail.com for evaluation.

References

- [1] European Commission (2021). Standard briefing slides for experts: Horizon Europe.
https://ec.europa.eu/info/funding-tenders/opportunities/docs/2021-2027/experts/standard-briefing-slides-for-experts_he_en.pdf
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- [3] European Commission (2025). Horizon Europe Work Programme 2026: European Research Council.
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- [4] FFG, Austrian Research Promotion Agency (2026). Funding rates in Horizon Europe.
https://www.ffg.at/en/europe/heu/legal-financial/theme_funding-rates
- [5] European Research Council (2025). Changes to the 2026 and 2027 Work Programmes.
<https://erc.europa.eu/news-events/news/changes-2026-and-2027-work-programmes>
- [6] National Science Foundation (2026). Merit review: Intellectual merit and broader impacts.
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